Fiscal Year:	FY 2010	**Task Last Updated:**	FY 08/06/2010
PI Name:	Rose, Raphael Ph.D.		
Project Title:	A Multi-Media, Computer-Based, Self-Directed, Autonomous, Stress and Anxiety-Management Countermeasure		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rose@psych.ucla.edu	**Fax:**	FY 310-825-9048
PI Organization Type:	UNIVERSITY	**Phone:**	310-825-9048
Organization Name:	University of California, Los Angeles		
PI Address 1:	Department of Psychology		
PI Address 2:	Box 951563, 1285 Franz Hall		
PI Web Page:			
City:	Los Angeles	**State:**	CA
Zip Code:	90095-1563	**Congressional District:**	33
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Crew Health NNJ07ZSA002N
Start Date:	07/01/2008	**End Date:**	06/30/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Buckey, Jay (Dartmouth College) Cartreine (nee Carter), James (Beth Israel Deaconess Medical Center) Craske, Michelle (University of California, Los Angeles) Beven, Gary (NASA JSC) Glover, Dorie (Dorothy) (University of California, Los Angeles)		
Grant/Contract No.:	NCC 9-58-NBPF01605		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Stress and anxiety-related problems before, during and after space missions can seriously compromise crew efficiency and safety. The Institute of Medicine report, of Earth-analog environments (e.g., Antarctica) suggests a prevalence rate of serious behavioral problems of up to 13% per person per year, highlighting the potential behavioral problems that could occur on a 3-year mission. Further, the potential severity of this problem is highlighted by Bioastraonautics Roadmap Risk #25 - human performance failure due to mood alterations such as depression, anxiety, or other psychiatric and cognitive problems - a Priority 1 risk for ISS and Mars. Our project has two specific aims. The first aim is to develop a self-directed, autonomous, interactive multimedia program to train astronauts how to recognize, assess, detect, prevent, and manage stress on extended spaceflights. The countermeasure would be principally used in pre-flight training, with booster sessions or interventions available on orbit as needed. In this model of implementation, the skills learned prior to a mission would help to "inoculate" fliers to severe stress reactions thereby enhancing flight-task performance and crew health, safety and efficiency, and build resilience. The countermeasure, called "Stress Management and Resilience Training for Optimal Performance on long-duration mission" or SMART-OP, will incorporate empirically supported stress and anxiety management strategies delivered by interactive media in the NSBRI-supported Virtual Space Station suite of psychosocial support programs. The second specific aim is to evaluate the efficacy of the program with sample of Navy Pilots (who are similar to astronauts) in a randomized controlled trial. The final product will be a computer-based stress management program that could be used both by flight crews and ground-personnel when needed both during training and during missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Stress Management and Resilience Training for Optimal Performance (SMART-OP), is a computer-based, self-guided multimedia, interactive, evidenced-based stress management program. At the end of this study, we will not only have developed an evidenced-based countermeasure with potential value to the space program and US Navy, but also with some modifications and revisions, to the broader American public. Stress-related health and mental problems and work-related stress and burnout are among the most common and costly in the country. Stress also plays a significant etiological role in the onset of many physical and mental health disorders. The Stress Diathesis Hypothesis states that stress and a biological predisposition are responsible for the onset of most mental health problems. After efficacy studies are complete, a potential next step would be larger-scale effectiveness studies to further identify in which sub-populations and user settings (home, clinic, etc.) it is most useful. With slight modifications, in addition to use in other operational environments like submarines, Antarctic research stations, commercial ships, and oil rigs. The program could be used with those who work in operationally-demanding professions (e.g., law enforcement, firefighters, emergency first responders, surgeons, and other high-stress professions). Ultimately, the program may have benefit in helping to prevent mental health disorders in the future by providing easy access to stress management skills that may help mitigate the onset of mental health disorders. SMART-OP has the potential for widespread dissemination via Internet to provide stress-management training to individuals worldwide.
Task Progress:	Intro: SMART-OP (Stress Management and Resilience Training for Optimal Performance on Long-duration Missions) is a stress management training program based on cognitive-behavioral principles. It teaches stress management skills in three main domains: thoughts, feelings and actions. SMART-OP will be tested for efficacy in US Navy Pilots. Program Design: We designed a 6-session training program and began production. The 6-session format was based on a review of stress management literature indicating that amount of time needed for proper effect sizes. The program is designed to introduce users to skills in each of the three domains (referenced above) during each "computer session," and then practice those exercises with aid of usb flashdrive at home and in-between sessions. We completed a SMART-OP Design Document. Program Content: We evaluated third party software for relaxation and biofeedback training exercises and chose two programs that will be incorporated to teach those skills in SMART-OP. The chosen programs will be adapted for use in SMART-OP and utilize a game-like interface to train users how to relax and have been used successfully with other military personnel. We evaluated and chose a third party software for compartmentalization and cognitive flexibility training exercises that will train the user to attend to multiple tasks while being distracted by visual and auditory stimuli. We created a test bank of stress-related questions for use in a knowledge challenge task. We wrote cognitive restructuring exercises and introduction and briefing scripts for the SMART-OP coach. We contacted long-duration astronauts and support staff, and US military personnel to ask for participation in videos and interviews, and filmed three interviews. We also interviewed various experts using structured interview based on stressful scenarios. The SMART-OP content document was incorporated into Astronaut Candidate (ASCAN) training by NASA behavioral health support team. We made the SMART-OP content document into a how to manual and stand alone publication and submitted it for publication. Content evaluation: The program was presented at an annual meeting of the Behavioral Health and Performance group and at an International Conference on Research in Space Analog Environments. We've had discussions with astronauts and US Navy and Marine personnel to review our work and approach. We reviewed the program's content with Dr. Donald Meichenbaum, Stress Inoculation Training creator and made modifications based on their feedback. We received unanimous feedback that SMART-OP will ultimately be most useful if it is incorporated into standard pre-flight/mission training and the importance that the program gets good support from those in authority to ensure proper "buy-in." The design of the program appears to be engaging and has received positive comments for its look and feel.
Bibliography Type:	Description: (Last Updated: 02/11/2021)
Awards	Buckey JC Jr. "Laurence R. Young Space Biomedical Research Award, February 2010." Feb-2010
Awards	Rose RD. "Distinguished Faculty Teaching Award - UCLA Psychology Department, June 2009." Jun-2009
Books/Book Chapters	Rose RD, Buckey JC, Craske MG, Cartreine JA. "Stress management and resilience training for optimal performance on long-duration space flight." Berkeley, Calif. : University of California Press, in press, 2010., May-2010